Evaluation of Recommendation Systems

CP4285 Modern Recommendation Systems — Week 03

CP4285 Instruction Team

25 Aug 2026

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soc-n.us/cp4285-t2610-w03

01 Announcements and Context

CHANGELOG

- **25 Aug 2026** — Added the Week 02 Canvas Voices review and guardrail discussion.
- **25 Aug 2026** — Added the Week 04 pre-lecture activity and linked preview image.

WEEK 02 RECAP

Three Routes to a Recommendation

Collaborative filtering (CF; last week)

Evidence
User–item interactions

Mechanism
Borrow from similar users or items

Watch
Thin local overlap can leave a neighbourhood silent.

Content-based filtering (CBF)

Evidence
Explicit item, user, or context features

IR mechanism
Encode and match genres, tags, and text

Watch
Feature coverage and encoding quality set the limit.

Latent-factor model (LFM)

Evidence
User–item interactions

Mechanism
Learn shared hidden coordinates from many interactions

Watch
A factor is a learned pattern, not a personal fact.

Worked TF–IDF: Two Titles, One Interest Profile

User interest: `animated family comedy`
A: *Toy Story* · Animation · Children's · Comedy **B:** *The Lego Movie* · Animation · Family · Comedy

Vocabulary coordinates

term	IDF	U	A	B
animated	1.8	1	0	0
animation	1.8	0	1	1
family	1.4	1	0	1
children	1.4	0	1	0
comedy	1.2	1	1	1

`animated` ≠ `animation`; `family` ≠ `children`.

Cosine = dot product ÷ (two vector norms)

$$\cos(\mathbf{u},\mathbf{t}) = \frac{\mathbf{u}\cdot\mathbf{t}}{|\mathbf{u}|\,|\mathbf{t}|} \quad \cos(\mathbf{u},\mathbf{t}) = \frac{\mathbf{u}\cdot\mathbf{t}}{|\mathbf{u}|\,|\mathbf{t}|}$$

A · Toy Story — 0.217
dot: 1.2 × 1.2 = 1.44
norms: ‖U‖ = 2.577; ‖A‖ = 2.577
1.44 ÷ (2.577 × 2.577)

B · The Lego Movie — 0.512
dot: 1.4² + 1.2² = 3.40
norms: ‖U‖ = 2.577; ‖B‖ = 2.577
3.40 ÷ (2.577 × 2.577)

Sparse and Dense Make Different Trade-offs

Question	TF–IDF	Dense word embedding
What is represented?	Exact vocabulary-term evidence	Learned continuous coordinates
Main strength	Transparent lexical matching	Can generalise across related usage
Main limitation	Synonyms remain separate terms	Learned proximity can be opaque or misleading
Typical score	Cosine similarity over sparse vectors	Cosine/dot-product similarity over dense vectors

Bridge: *IR turns* ***text into comparable vectors****. LFM will turn* ***interaction history into comparable user and item vectors****.*

A Dot Product Turns Factors into a Score

$$\hat{r_{ui}} = \mathbf{u}_u^\top \mathbf{v}_i$$

$$\hat{r_{ui}} = \mathbf{u}_u^\top \mathbf{v}_i$$

User vector

$$\mathbf{u}_u = [u_{u1}, u_{u2}, \ldots, u_{uk}]$$

$$\mathbf{u}_u = [u_{u1}, u_{u2}, \ldots, u_{uk}]$$

A learned location in the model's preference space.

Item vector

$$\mathbf{v}_i = [v_{i1}, v_{i2}, \ldots, v_{ik}]$$

$$\mathbf{v}_i = [v_{i1}, v_{i2}, \ldots, v_{ik}]$$

How an item aligns with the learned interaction patterns.

The dot product produces a **model score**. A production recommender can still filter, diversify, or re-rank the resulting candidates.

Recap: Yesterday’s deadlines

MON, 24 AUG · 23:59 SGT

- ✓ **Essay e1**
- ✓ **Mini-team formation**
- ✓ **Personal ID**
- ✓ **Pre-lecture exercise**



Essay `e1` Peer Review

YOUR TASK

TA Sahaj  will manually assign you another student's essay that is on a different recommendation platform.

1. Read the assigned essay.
2. Review and critique that essay only for the card suit lens.

FOCUS OF CRITIQUE

Assess the quality of the essay's **card suit** lens' quality.

- How well does it match the recommendation surface?
- How useful is the insight?

The two marks for the **peer review** are for the completion of your peer review and its quality.

Your essay's quality will be assessed by the eight marks given by the instruction staff.

Due: Mon, 31 Aug 2026 · 23:59 SGT

> **AI-use note.**As per the `e1` specification, remember AI should not be used in completing the peer review.



LAST WEEK PRE-LECTURE EXERCISE

What Should Evaluation Be Responsible For?

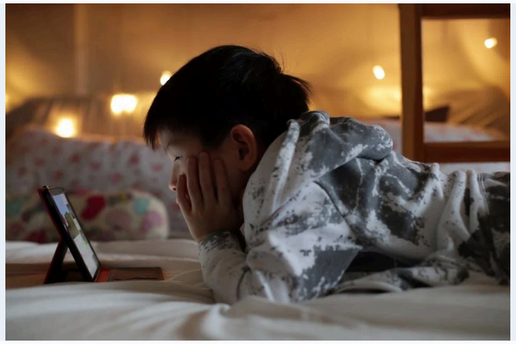


PHOTO: ST FILE

Read: “S’pore studying restrictions on direct messaging, auto-play on social media to protect kids”

Canvas discussion board · post one individual comment before **Mon, 23:59 SGT**

> **A recommender may be evaluated using whether people watch, click, or continue interacting with suggested content. What is one responsibility that evaluation should include for younger users, beyond predicting continued engagement?**

In **120–160 words**, choose and name **one card-suit lens**, cite one article detail, and propose one piece of evidence, safeguard, or user control the platform should examine.

No classmate reply this week. Today: protocols, ranking metrics, and what a score leaves out.

LAST WEEK PRE-LECTURE EXERCISE · CANVAS VOICES

Diamond and Clubs guardrails students proposed ♦ ♣



Evaluation makes a claim about success

> *A model can score user–item pairs. An evaluation protocol decides whether those scores support a useful, defensible decision.*

Note

Today: design the protocol, read ranking metrics, and ask what the metrics leave out.



🔑 Key Point: Evaluation Is More Than a Score

Defensible evaluation connects a protocol, a ranking metric, and the limits of the evidence.

1. Design
evaluation protocols

2. Compute
ranking metrics

3. Critique
metric selection

4. Explain
limitations of offline evaluation

🤞 **Ask what the recommender is being credited for—and what the evidence cannot establish.**



Bandit Time: What did the model learn?

A user has never clicked a recommended film. Which conclusion is justified?

A. The user dislikes the film.

B. The film was shown and ignored for a known reason.

C. The log alone cannot separate preference from exposure or interface effects.

D. The latent-factor model has no prediction for this user.

00:30

▶ Start Timer

TempoQuiz debrief: absence is not a label

Tempting shortcut

“No click” means dislike, so treat it as a negative label.

Evaluation-aware view

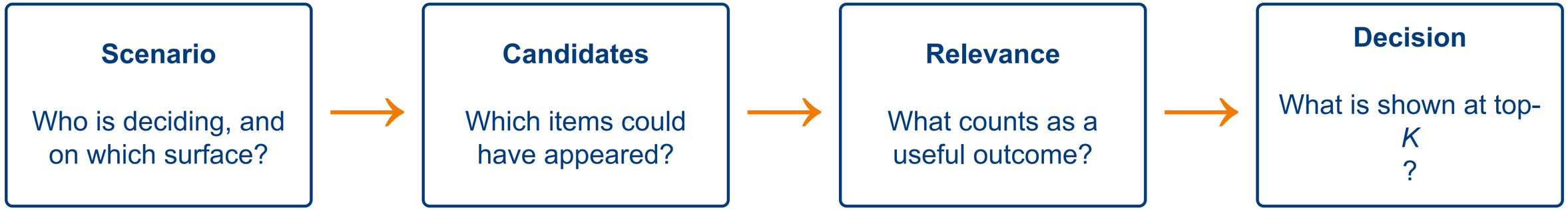
No click may mean no exposure, poor placement, wrong time, inattention, or lack of interest. The protocol must say which interpretation is defensible.



02 Evaluation Design



The evaluation contract



> *Do not choose a metric until these four terms are explicit.*

Bandit Time: Which split supports this claim?

You claim a news recommender will work next week. Which test design best supports that claim?

A. Train and test on the same logged clicks.

B. Randomly split all interactions, including future events and features.

C. Train on earlier interactions; tune on a later validation period; test on an untouched future period.

D. Choose the split after seeing which one gives the highest NDCG.

00:30

▶ Start Timer

Training, validation, and testing serve different decisions

Train

Fit the recommender using only information available at the training cutoff.

Validate

Choose hyperparameters, thresholds, and model variants without touching the final test period.

Test once

Estimate performance of the chosen system on an untouched period; do not tune again after seeing it.



03 Ranked Lists, Not Just Scores



👥 Complete the Metric Lens Table

In pairs, complete the four orange cells. Be ready to justify one choice from the ranking behaviour it measures.

Metric / signal	Question answered	Useful when
Precision@K	Of the *K* shown items, how many were relevant?	_____
Recall@K	How much of the relevant set was retrieved?	_____
Mean Average Precision (MAP)	Are relevant items repeatedly placed early?	_____
Normalised Discounted Cumulative Gain (NDCG)	Are the most valuable items at the top?	_____



Offline ranking metrics

Metric / signal	Question answered	Useful when
Precision@K	Of the shown *K*, how many were relevant?	Visible space is scarce.
Recall@K	How much relevant content was retrieved?	Missing useful items is costly.
MAP	Are binary-relevant items repeatedly early?	Several relevant items may matter.
NDCG	Are higher-value items early?	Relevance is graded; rank matters.

🤔 **To think about**

How do precision and recall evolve as *k* increases? When is P@K or R@K more useful as an evaluation lens?



👥 Which recommender wins?

For a five-item music shelf, choose the better recommender and defend one metric.

A: early high-value hit

1. ★★★★★ 2. ○ 3. ○ 4. ★★ 5. ○

B: more binary hits

1. ★★ 2. ★★ 3. ○ 4. ★★ 5. ○

03:00

▶ Start Timer

In groups of 3–4: name the surface assumption, the metric, and one reason the other system might still be preferred.

Worked MAP@5: binary relevance

A · 1. ★★★★★ · 2. ○ · 3. ○ · 4. ★★ · 5. ○

B · 1. ★★ · 2. ★★ · 3. ○ · 4. ★★ · 5. ○

For MAP, ★★ and ★★★★★ are both binary hits.

$$AP@5 = \frac{1}{R} \sum_{i=1}^{5} P@i \times rel_i$$

A: hits at ranks 1 and 4

Relevant ranks: **1, 4**

$$AP@5 = \frac{P@1 + P@4}{2} = \frac{1 + \frac{2}{4}}{2} = \mathbf{.750}$$

B: hits at ranks 1, 2, and 4

Relevant ranks: **1, 2, 4**

$$AP@5 = \frac{P@1 + P@2 + P@4}{3} = \frac{1 + 1 + \frac{3}{4}}{3} = \mathbf{.917}$$

🤔 **Would graded relevance change the result?** 📶

Yes—but then this is no longer standard MAP. A different relevance threshold changes AP; inserting rating grades directly creates a different graded metric and may favour A.

Worked NDCG@5: graded relevance

A · 1. ★★★★★ · 2. ○ · 3. ○ · 4. ★★ · 5. ○

B · 1. ★★ · 2. ★★ · 3. ○ · 4. ★★ · 5. ○

Grades: ○ = 0, ★★ = 2, ★★★★★ = 5. The gain term preserves their difference.

$$\text{NDCG@5} = \frac{\text{DCG@5}}{\text{IDCG@5}}, \qquad \text{DCG@5} = \sum_{i=1}^{5} \frac{2^{\text{rel}_i} - 1}{\log_2(i+1)}$$

A: ★★★★★ at 1; ★★ at 4

$$\text{DCG} = 31 + \frac{3}{\log_2 5} = 32.292$$

$$\text{IDCG} = 31 + \frac{3}{\log_2 3} = 32.893$$

$$\text{nDCG@5} = \mathbf{.982}$$

B: ★★ at 1, 2, and 4

$$\text{DCG} = 3 + \frac{3}{\log_2 3} + \frac{3}{\log_2 5} = 6.185$$

$$\text{IDCG} = 3 + \frac{3}{\log_2 3} + \frac{3}{2} = 6.393$$

$$\text{nDCG@5} = \mathbf{.968}$$

Note

IDCG == Idealised Discounted Cumulative Gain. An oracular value that simulates the best possible outcome

Click-through rate (CTR): an online metric

Calculation

$$\text{CTR} = \frac{\text{clicks}}{\text{impressions}} \times 100\%$$

Impression: a visible recommendation opportunity.
Click: an opening or action that your instrumentation counts.

Worked example

$$\frac{28 \text{ clicks}}{400 \text{ impressions}} \times 100\% = \mathbf{7\%}$$

CTR asks whether people acted after seeing the recommendation. It is measured online, after a deployed surface creates impressions.

🤔 To think about

How do you apply CTR to our two use cases? What assumptions should you make?

🔑 Key Point: A Metric Orders Priorities

A system can be better at the top, across many relevant items, or at avoiding wasted slots.

Top of the list

Reward especially valuable items that appear early.

Repeated relevance

Reward several relevant items placed early.

Wasted slots

Penalise irrelevant items in scarce visible space.

🤞 **Your metric makes one of these priorities count most.**



Break — 5 minutes

☕ **BREAK · CTR DEPENDS ON THE SURFACE**

The same recommendation can earn different click-through rates because the surface changes the user's intent, the competing choices, and the meaning of an impression.

Paid search ads

6.64%

Overall CTR across 13,000 U.S. campaigns.

WordStream (2026)

Social traffic ads

1.71%

Average CTR for Facebook traffic campaigns.

WordStream (2025)

Email

Context matters

Audience, message, and timing shape the response.

Mailchimp (n.d.)



Rule: compare CTR only across like-for-like surfaces, placements, audiences, and impression definitions.

Sources (APA). WordStream. (2026). *PPC benchmarks*. Source · WordStream. (2025). *Facebook ads benchmarks 2025*. Source · Mailchimp. (n.d.). *Email marketing benchmarks and industry statistics*. Source

04 Beyond Accuracy



Accuracy is a proxy, not a verdict

> ⚠ **Warning**
>
> An observed click can be a useful signal without being a complete account of benefit, consent, satisfaction, or long-term welfare.



Same precision, different catalogue

System P: popular repeat

Every user sees familiar blockbusters. Precision@5 is high, but the catalogue is barely touched.

Likely risk: repeated exposure and narrow discovery.

System D: relevant discovery

Each user sees useful, less familiar options across the catalogue. Precision@5 is the same.

Potential value: novelty, diversity, and long-tail exposure.



👥 Metric design clinic: choose the guardrails

Your group is assigned one card lens. Design a minimal evaluation plan for a recommendation surface.

♣ Clubs — adversarial / shill

Can coordinated activity manipulate the list? Consider attack impact, anomaly detection, and stability.

♥ Hearts — social

Does social evidence help without unwanted exposure or conformity? Consider privacy, welfare, and subgroup outcomes.

♦ Diamonds — novelty seeker

Does the list enable discovery? Consider novelty, diversity, serendipity, and catalogue coverage.

♠ Spades — loyalist / metadata-driven

Are important requirements stable and visible? Consider constraints, consistency, calibration, and explanation.

03:00

▶ Start Timer

Name: (1) one primary metric; (2) two guardrails; and (3) one evidence gap that offline logs cannot resolve. Prepare a two-sentence report.



Debrief: metric choices distribute attention

Lens	Possible primary measure	Example guardrail
♣	Ranking utility under attack	Attack impact / stability
♥	Task completion or satisfaction	Privacy, opt-outs, welfare signal
♦	NDCG or utility	Novelty, diversity, coverage
♠	Constraint-satisfied relevance	Consistency / explanation quality



Evaluation criteria map

USEFUL NOW
Relevance & utility
Precision, recall, MAP, NDCG, task completion

BEYOND RELEVANCE
Engagement, exposure & discovery
CTR, dwell time, exposure distribution, coverage, intra-list diversity, novelty, serendipity

SAFE AND DEPENDABLE
Trust & resilience
Confidence, robustness, stability, scalability

🤔 To think about

Where should the computation budget sit—candidate generation, ranking, or re-ranking? What end-to-end latency can this surface tolerate before a useful recommendation arrives too late?

ⓘ Note

Measure the three ideas separately. *Dwell time* captures depth after an interaction; *exposure distribution* records who and what receives impressions; *intra-list diversity* asks how varied one ranked list is. None is interchangeable with relevance.

Beyond relevance: measure what receives attention

Engagement

CTR · dwell time

Did people act after a visible recommendation opportunity?

Exposure

Impression share · coverage

Who and what receives a place in the ranked list?

Discovery

Diversity · novelty · serendipity

Does the list expand useful options beyond familiar repeats?

Note

These measures answer different questions. A system can earn clicks while concentrating exposure, or improve discovery while leaving CTR unchanged.

Intra List Diversity (ILD): how varied is one ranked list?

A three-item recommendation list

Item A, Item B, Item C

Use a content representation to estimate each pair's cosine similarity.

A–B	A–C	B–C
similarity = 0.80	similarity = 0.20	similarity = 0.30
distance = 0.20	distance = 0.80	distance = 0.70

Average pairwise dissimilarity

$$\mathrm{ILD@3} = \frac{0.20+0.80+0.70}{3} = 0.57$$

Interpretation: this list has moderate intra-list diversity on a 0–1 distance scale. A higher score means the items are less alike *under the chosen representation*.

🤔 To think about

What item representation would make two recommendations look more—or less—similar: genre labels, text embeddings, creator, or user interaction patterns? What is the complexity of this calculation? Is it important to consider in practical service?

Safe and dependable: measure whether the system holds up

Trust

Calibration · explanation coverage

Are confidence and explanations proportionate to what the system knows?

Resilience

Attack impact · ranking stability

How much does the list change under manipulation or small perturbations?

Scalability

p95 latency · throughput

Can the surface respond reliably under its expected traffic and catalogue size?

> ⚠ **Warning**
>
> There is no universal safety score. Select operational and resilience checks that match the surface, the users affected, and the failure mode at stake.



95% Percent Latency (p95): Protect the slow tail

20 sorted recommendation-response latencies (ms)

120 · 125 · 128 · 130 · 135 · 140 · 145 · 148 · 150 · 152

155 · 160 · 165 · 170 · 175 · 180 · 190 · 205 · **240** · 800

Use the nearest-rank convention for this worked example.

Find the 95th percentile

$$\text{rank} = \lceil 0.95 \times 20 \rceil = 19$$

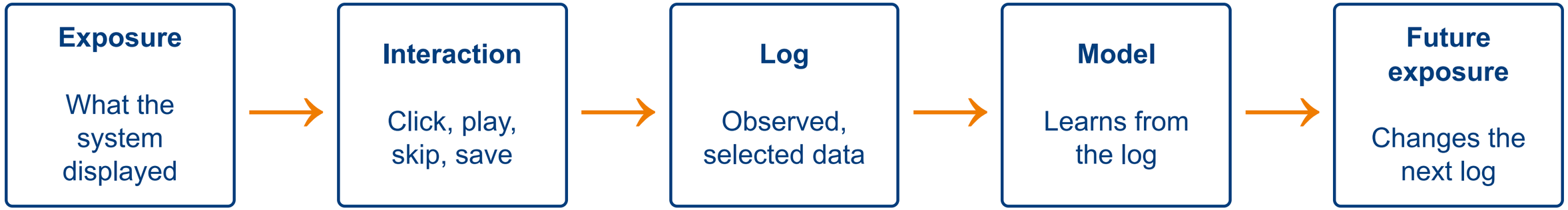
$$p95 = L_{19} = 240 \text{ ms}$$

Interpretation: 95% of requests completed in 240 ms or less; the slowest 5% took longer. The single 800 ms request still matters, but it is not p95.

🤔 **To think about**

What are appropriate interventions to address long latency as a result of a high p95?

The feedback loop hidden in a log



⚠ Warning

Unobserved is not random. The system's own past choices influence what can be learned and evaluated.

Four offline failure modes

MNAR / selection bias

Logged feedback reflects what users chose and what the system exposed. Mitigate with temporal realism, exposure data, and careful claims.

Temporal drift

Preferences and catalogues change. Use forward-looking splits and report the evaluation window.

Leakage

Future features or test-set tuning inflate results. Enforce a feature cutoff and final untouched test period.

Cold start and long tail

Averages can hide new users, new items, and rare items. Report slices and coverage, not just one aggregate score.



Offline → online → user impact

Offline benchmark

Compare candidates under a declared protocol; inspect coverage and intra-list diversity

→

Guarded online test

Randomise exposure; monitor CTR, dwell time, exposure distribution, and safeguards

→

User / product impact

Interpret retention, satisfaction, complaints, and long-term effects



Summary

🔑 Evaluation choices define what counts as a good recommendation.

1 · Begin with the claim
Name the surface, decision, candidates, and relevant outcome before choosing a split or metric.

2 · Judge ranked lists
Match Precision@K, Recall@K, MAP, or NDCG to the ranking behaviour that matters.

3 · Add guardrails
Pair utility with discovery, exposure, trust, robustness, and constraint checks that fit the card lens.

4 · Keep claims proportional
Offline logs are a simulation: test forward in time, report evidence gaps, then validate with guarded online evidence.

A metric orders priorities; an evaluation plan makes those priorities defensible.



NEXT WEEK PRE-LECTURE EXERCISE

What Does a Missing Interaction Mean?

Canvas discussion board · post before Mon, 31 Aug 23:59 SGT

Post one concise response; then reply constructively to one classmate.

> **A user has never clicked a recommended item. Give two plausible interpretations: one about preference and one about exposure or the interface.**

Next week: neural recommenders can learn richer interaction patterns, but they still learn only from the evidence the system records.



NEXT WEEK PREVIEW — WEEK 04

Neural Recommendation Models

A neural recommender can learn richer patterns from interactions. That does not make an unclicked item self-explanatory.

In Week 4:

- compare neural and latent-factor recommenders;
- examine deep ranking and representation learning; and
- ask what model complexity can—and cannot—explain.



Illustration: Google DeepMind / Pexels